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MENOPUR® 75IU

Powder and Solvent for Solution for Injection

COMPOSITION

MENOPUR® 75IU: The reconstituted solution (1 mL) contains 75 IU of highly purified menotrophin (human menopausal gonadotrophin, hMG). Each vial with powder contains hMG corresponding to 75 IU follicle stimulating hormone (FSH) bioactivity and 75 IU luteinizing hormone (LH) bioactivity.

Origin of the active substances:

Menotrophin (human menopausal gonadotrophin, hMG) originates from the urine of postmenopausal women. It contains follicle stimulating hormone (FSH), luteinising hormone (LH) and human chorionic gonadotrophin (hCG), providing FSH bioactivity and LH bioactivity in a 1:1 ratio.

hCG, a hormone naturally present in the urine of postmenopausal and pregnant women, is the primary contributor to the LH bioactivity. Extracted hCG from the urine of pregnant women may be used to balance the total LH bioactivity.

List of excipients:

Powder: lactose monohydrate, polysorbate 20, sodium hydroxide, hydrochloric acid

Solvent: sodium chloride, hydrochloric acid, water for injections.

Sodium content:

MENOPUR® contains less than 1 mmol sodium (23mg) per dose, so it is essentially 'sodium free'.

PHARMACEUTICAL DOSAGE FORM

Powder and solvent for solution for injection

INDICATIONS

MENOPUR® is indicated for the treatment of infertility in the following clinical situations:

Anovulation, including polycystic ovarian disease (PCOD), in women who have been unresponsive to treatment with clomiphene citrate.

Controlled ovarian hyperstimulation to induce the development of multiple follicles for assisted reproductive technologies (ART) (e.g. in vitro fertilisation/embryo transfer (IVF/ET), gamete intra-fallopian transfer (GIFT) and intracytoplasmic sperm injection (ICSI).

DOSAGE AND ADMINISTRATION

Treatment with MENOPUR® should be initiated under the supervision of a physician experienced in the treatment of fertility problems.

Method of administration

Dosage regimens described below are identical for subcutaneous (S.C.) and intramuscular (I.M.) administration.

MENOPUR® 75 IU are intended for subcutaneous (S.C.) or intramuscular (I.M.) injection after reconstitution with the solvent provided.

The powder should be reconstituted immediately prior to use. In order to avoid the injection of large volumes, up to 3 vials of the powder may be dissolved in 1 ml of the solvent provided.

Shaking should be avoided. The solution should not be used if it contains particles or if it is not clear.

Dosage

The inter-individual variation in the response of the ovaries to exogenous gonadotrophins is high. This makes it impossible to set a uniform dosage scheme. The dosage should, therefore, be adjusted individually depending on the ovarian response. MENOPUR® can be given alone or in combination with a gonadotrophin-releasing hormone (GnRH) agonist or antagonist. Recommendations about dosage and duration of treatment may change depending on the actual treatment protocol.

Women with anovulation (including PCOD):

The objective of MENOPUR® therapy is to develop a single Graafian follicle from which the oocyte will be liberated after the administration of human chorionic gonadotrophin (hCG).

MENOPUR® therapy should start within the initial 7 days of the menstrual cycle. The recommended initial dose of MENOPUR® is 75-150 IU daily, which should be maintained for at least 7 days. Based on clinical monitoring (including ovarian ultrasound alone or in combination with measurement of oestradiol levels) subsequent dosing should be adjusted according to individual patient response. Adjustments in dose should not be made more frequently than every 7 days. The recommended dose increment is 37.5 IU per adjustment, and should not exceed 75 IU. The maximum daily dose should not be higher than 225 IU. If a patient fails to respond adequately after 4 weeks of treatment, that cycle should be abandoned and the patient should recommence treatment at a higher starting dose than in the abandoned cycle.

When an optimal response is obtained, a single injection of 5,000 IU to 10,000 IU hCG should be given 1 day after the last MENOPUR® injection. The patient is recommended to have coitus on the day of and the day following hCG administration. Alternatively intrauterine insemination (IUI) may be performed. If an excessive response to MENOPUR® is obtained, treatment should be stopped and hCG withheld (see Special warning and precautions for use) and the patient should use a barrier method of contraception or refrain from having coitus until the next menstrual bleeding has started.

Women undergoing controlled ovarian hyperstimulation for multiple follicular development for assisted reproductive technologies (ART) (e.g. in vitro fertilization/embryo transfer (IVF/ET), gamete intra-fallopian transfer (GIFT) and intracytoplasmic sperm injection (ICSI)):

In a protocol using down-regulation with a GnRH agonist, MENOPUR® therapy should start approximately 2 weeks after the start of agonist treatment. In a protocol using down-regulation with a GnRH antagonist, MENOPUR® therapy should start on day 2 or 3 of the menstrual cycle. The recommended initial dose of MENOPUR® is 150-225 IU daily for at least the first 5 days of treatment. Based on clinical monitoring (including ovarian ultrasound alone or in combination with measurement of oestradiol levels) subsequent dosing should be adjusted according to individual patient response, and should not exceed more than 150 IU per adjustment. The maximum daily dose given should not be higher than 450 IU daily and in most cases dosing beyond 20 days is not recommended.

When a suitable number of follicles have reached an appropriate size a single injection of up to 10,000 IU hCG should be administered to induce final follicular maturation in preparation for oocyte retrieval. Patients should be followed closely for at least 2 weeks after hCG administration. If an excessive response to MENOPUR® is obtained treatment should be stopped and hCG withheld (see section Special warnings and precautions for use) and the patient should use a barrier method of contraception or refrain from having coitus until the next menstrual bleeding has started.

Renal/hepatic impairment

Patients with renal and hepatic impairment have not been included in clinical trials (see section Pharmacokinetics).

Paediatric population

There is no relevant use of MENOPUR® in the paediatric population.

Elderly

There is no relevant use of MENOPUR® in the elderly population.

CONTRAINDICATIONS

MENOPUR® is contraindicated in women who have:

- Tumours of the pituitary gland or hypothalamus
- Ovarian, uterine or mammary carcinoma
- Pregnancy and lactation
- Gynaecological haemorrhage of unknown aetiology
- Hypersensitivity to the active substance or any of the excipients used in the formulation (see section Incompatibilities)
- Ovarian cysts or enlarged ovaries not due to polycystic ovarian disease.

In the following situations treatment outcome is unlikely to be favourable, and therefore MENOPUR® should not be administered:

- Primary ovarian failure
- Malformation of sexual organs incompatible with pregnancy
- Fibroid tumours of the uterus incompatible with pregnancy

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

MENOPUR® is a potent gonadotrophic substance capable of causing mild to severe adverse reactions and should only be used by physicians who are thoroughly familiar with infertility problems and their management.

Gonadotrophin therapy requires a certain time commitment by physicians and supportive health professionals, and calls for monitoring of ovarian response with ultrasound, alone or in combination with measurement of serum oestradiol levels, on a regular basis. There is considerable inter-patient variability in response to menotrophin administration, with a poor response to menotrophin in some patients. The lowest effective dose in relation to the treatment objective should be used.

The first injection of MENOPUR® should be performed under direct medical supervision.

Before starting treatment, the couple's infertility should be assessed as appropriate and putative contraindications for pregnancy evaluated. In particular, patients should be evaluated for hypothyroidism, adrenocortical deficiency, hyperprolactinaemia and pituitary or hypothalamic tumours, and appropriate specific treatment given.

Patients undergoing stimulation of follicular growth, whether in the frame of a treatment for anovulatory infertility or ART procedures may experience ovarian enlargement or develop hyperstimulation. Adherence to recommended MENOPUR® dosage and regimen of administration, and careful monitoring of therapy will minimise the incidence of such events. Acute interpretation of the indices of follicle development and maturation requires a physician who is experienced in the interpretation of the relevant tests.

Ovarian Hyperstimulation Syndrome (OHSS)

OHSS is a medical event distinct from uncomplicated ovarian enlargement. OHSS is a syndrome that can manifest itself with increasing degrees of severity. It comprises marked ovarian enlargement, high serum sex steroids, and an increase in vascular permeability which can result in an accumulation of fluid in the peritoneal, pleural and, rarely, in the pericardial cavities.

The following symptoms may be observed in severe cases of OHSS: abdominal pain, abdominal distension, severe ovarian enlargement, weight gain, dyspnoea, oliguria and gastrointestinal symptoms including nausea, vomiting and diarrhoea. Clinical evaluation may reveal hypovolaemia, haemoconcentration, electrolyte imbalances, ascites, haemoperitoneum, pleural effusions, hydrothorax, acute pulmonary distress, and thromboembolic events.

Excessive ovarian response to gonadotrophin treatment seldom gives rise to OHSS unless hCG is administered to trigger ovulation. Therefore, in cases of ovarian hyperstimulation it is prudent to withhold hCG and advise the patient to refrain from coitus or to use barrier methods for at least 4 days. OHSS may progress rapidly (within 24 hours to several days) to become a serious medical event, therefore patients should be followed for at least two weeks after the hCG administration.

Adherence to recommended MENOPUR® dosage, regimen of administration and careful monitoring of therapy will minimise the incidence of ovarian hyperstimulation and multiple pregnancy (see sections Dosage and administration and Undesirable effects). In ART, aspiration of all follicles prior to ovulation may reduce the occurrence of hyperstimulation.

OHSS may be more severe and more protracted if pregnancy occurs. Most often, OHSS occurs after hormonal treatment has been discontinued and reaches its maximum severity at about seven to ten days following treatment. Usually, OHSS resolves spontaneously with the onset of menses.

If severe OHSS occurs, gonadotrophin treatment should be stopped if still ongoing, the patient hospitalised and specific therapy for OHSS started.

This syndrome occurs with higher incidence in patients with polycystic ovarian disease.

Multiple pregnancy

Multiple pregnancy, especially high order, carries an increased risk of adverse maternal and perinatal outcomes.

In patients undergoing ovulation induction with gonadotrophins, the incidence of multiple pregnancies is increased compared with natural conception. The majority of multiple conceptions are twins. To minimise the risk of multiple pregnancy, careful monitoring of ovarian response is recommended.

In patients undergoing ART procedures the risk of multiple pregnancy is related mainly to the number of embryos replaced, their quality and the age of the patient.

The patient should be advised of the potential risk of multiple births before starting treatment.

Pregnancy loss

The incidence of pregnancy loss by miscarriage or abortion is higher in patients undergoing stimulation of follicular growth for ART procedures than in the normal population.

Ectopic pregnancy

Women with a history of tubal disease are at risk of ectopic pregnancy, whether the pregnancy is obtained by spontaneous conception or with fertility treatment. The prevalence of ectopic pregnancy after IVF has been reported to be 2 to 5%, as compared to 1 to 1.5% in the general population.

Reproductive system neoplasms

There have been reports of ovarian and other reproductive system neoplasms, both benign and malignant, in women who have undergone multiple drug regimens for infertility treatment. It is not yet established if treatment with gonadotrophins increases the baseline risk of these tumours in infertile women.

Congenital malformation

The prevalence of congenital malformations after ART may be slightly higher than after spontaneous conceptions. This is thought to be due to differences in parental characteristics (e.g. maternal age, sperm characteristics) and multiple pregnancies.

Thromboembolic events

Women with generally recognised risk factors for thromboembolic events, such as personal or family history, severe obesity (Body Mass Index > 30 kg/m²) or thrombophilia may have an increased risk of venous or arterial thromboembolic events, during or following treatment with gonadotrophins. In these women, the benefits of gonadotrophin administration need to be weighed against the risks. It should be noted however, that pregnancy itself also carries an increased risk of thromboembolic events.

PREGNANCY AND BREASTFEEDING

Fertility

MENOPUR® is indicated for use in infertility (see section Indication)

Pregnancy

MENOPUR® is contraindicated in women who are pregnant. (see section Contraindication)

There are no or limited amount of data from the use of menotrophins in pregnant women. No animal studies have been carried out to evaluate the effect of MENOPUR® during pregnancy.

Breastfeeding

MENOPUR® is contraindicated in women who are breastfeeding. (see section Contraindication)

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

No drug-drug interaction studies have been performed with MENOPUR® in humans.

Although there is no controlled clinical experience, it is expected that the concomitant use of MENOPUR® and clomiphene citrate may enhance the follicular response. When using GnRH agonist for pituitary desensitisation, a higher dose of MENOPUR® may be necessary to achieve adequate follicular response.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on the ability to drive and use machines have been performed. However, MENOPUR® is unlikely to have influence on the patient's ability to drive and use machines.

INCOMPATIBILITIES

MENOPUR® should not be administered in the same injection with other products.

UNDESIRABLE EFFECTS

The most frequently reported adverse drug reactions (ADR) during treatment with MENOPUR® in clinical trials are Ovarian Hyperstimulation Syndrome (OHSS), headache, pelvic pain, pelvic discomfort, abdominal pain, abdominal distension, nausea and injection site pain. None of these ADRs have been reported with an incidence rate of more than 5%.

The table below displays the main ADRs in women treated with MENOPUR® in clinical trials distributed by system organ classes (SOCs) and frequency. Further, the ADRs seen during post-marketing experience are mentioned with unknown frequency.

System Organ Class	Common (> 1/100 to < 1/10)	Uncommon (> 1/1,000 to < 1/100)	Rare (> 1/10,000 to < 1/1,000)	Unknown
Eye disorders				Visual impairment
Gastrointestinal disorders	Abdominal pain, Abdominal distension, Nausea	Vomiting, Abdominal discomfort, Diarrhoea		
General disorders and administration site condition	Injection site reactions ^a	Fatigue		Pyrexia, Malaise
Immune system disorders				Hypersensitivity reactions ^b
Investigations				Weight increased
Musculoskeletal & connective tissue disorders				Musculoskeletal pain ^c
Nervous system disorders	Headache	Dizziness		
Reproductive system disorders	OHSS ^d , Pelvic pain ^e	Ovarian cyst, Breast complaints ^f		Ovarian torsion ^d
Skin and subcutaneous tissue disorders			Acne, Rash	Pruritus, Urticaria
Vascular Disorders		Hot flush		Thromboembolism ^d

^a Most frequently reported injection site reaction was injection site pain.

^b Cases of localised or generalised allergic reactions, including anaphylactic reaction, along with associated symptomatology have been reported rarely.

^c Musculoskeletal pain includes arthralgia, back pain, neck pain and pain in extremities.

^d Gastrointestinal symptoms associated with OHSS such as abdominal distension and discomfort, nausea, vomiting, diarrhoea have been reported with MENOPUR[®] in clinical trials. In cases of severe OHSS ascites and pelvic fluid collection, pleural effusion, dyspnoea, oliguria, thromboembolic events and ovarian torsion have been reported as rare complications.

^e Pelvic pain includes ovarian pain and adnexa uteri pain.

^f Breast complaints include breast pain, breast tenderness, breast discomfort, nipple pain and breast swelling.

PHARMACODYNAMICS

Pharmacotherapeutic group: Gonadotrophins

ATC code: G03G A02

MENOPUR, administered for 7 to 20 days, produces ovarian follicular growth and maturation in women who do not have primary ovarian failure. Treatment with MENOPUR in most instances results only in follicular growth and maturation. When sufficient follicular maturation has occurred, hCG must be given to induce ovulation.

FSH bioactivity is the primary driver of follicular recruitment and growth in early folliculogenesis, while LH bioactivity is important for ovarian steroidogenesis and is involved in the physiological events leading to the development of a competent pre-ovulatory follicle. Follicular growth can be stimulated by FSH bioactivity in the total absence of LH bioactivity, but the resulting follicles develop abnormally and are associated with low oestradiol levels and inability to luteinize to a normal ovulatory stimulus.

Menotrophin, which contains both FSH bioactivity and hCG-driven LH bioactivity, induces ovarian follicular growth and development as well as gonadal steroid production in women who do not have primary ovarian failure.

In line with the action of LH bioactivity in enhancing steroidogenesis, oestradiol levels associated with treatment with MENOPUR[®] are higher than with recombinant FSH preparations in downregulated IVF/ICSI cycles. This issue should be considered when monitoring patients' response based on oestradiol levels. The difference in oestradiol levels is not found when using low-dose ovulation induction protocols in anovulatory patients.

PHARMACOKINETICS

The pharmacokinetic profile of the FSH in MENOPUR[®] has been documented. After 7 days of repeated dosing with 150 IU MENOPUR[®] in downregulated healthy female volunteers, maximum plasma FSH concentrations C_{max} (baseline-corrected) (mean ± SD) were 8.9 ± 3.5 IU/L and 8.5 ± 3.2 IU/L for the SC and IM administration, respectively. Maximum FSH concentrations were reached (T_{max}) within 7 hours for both routes of administration. After repeated administration, FSH was eliminated with a half-life (T_{1/2}) (mean ± SD) of 30 ± 11 hours and 27 ± 9 hours for the SC and IM administration, respectively. Although the individual LH concentration versus time curves show an increase in the LH concentration after dosing with MENOPUR[®], the data available were too sparse to be subjected to a pharmacokinetic analysis.

Menotrophin is mainly excreted through the kidneys with a contribution through the liver.

The pharmacokinetics of MENOPUR[®] in patients with renal or hepatic impairment has not been investigated.

PRECLINICAL SAFETY DATA

Non-clinical data reveal no special hazard for humans, which is not known from the extensive clinical experience.

Reproduction toxicity studies have not been carried out to evaluate the effects of MENOPUR[®] during pregnancy or postpartum as MENOPUR[®] is not indicated during these periods.

MENOPUR[®] consists of naturally occurring hormones and should be expected to be non-genotoxic. Carcinogenicity studies have not been carried out as the indication is for short-term treatment.

OVERDOSAGE

The effects of an overdose is unknown, nevertheless one could expect ovarian hyperstimulation syndrome to occur.

SHELF-LIFE

2 years.

For immediate and single use following reconstitution.

STORAGE

Do not store above 30 °C. Do not freeze. Store in the original container in order to protect from light.

PACKING SIZES

MENOPUR® 75IU is available in the following containers and pack sizes:

Powder: 4 mL colourless glass (Type I) vials with rubber stopper closed with a cap.

Solvent: 1 mL colourless glass (Type I) ampoule.

The product is supplied in packs of 5 or 10 vials with the corresponding number of solvent ampoules.

INSTRUCTIONS FOR USE AND HANDLING

The powder should only be reconstituted with the solvent provided in the package.

Attach a reconstitution needle to the syringe. Break off the top of the ampoule containing the solvent. Withdraw the entire content from the ampoule with solvent and inject the total contents into the vial containing the powder. When the solvent needle pierces the vial there is a slight vacuum inside the vial which may lead to the powder completely dissolving with only a few drops of solvent. This may give the false impression that there was no powder in the vial, please be sure to add the entire solvent and proceed to the next step of administration. Shaking should be avoided.

If needed, the solution can be drawn up into the syringe again to transfer it to the next vial with powder until the prescribed dose has been reached. Up to three powder vials can be dissolved with one ampoule of solvent.

When the prescribed dose has been reached, draw up the mixed solution from the vial into the syringe, change to a hypodermic needle and administer immediately.

The reconstituted solution should not be administered if it contains particles or is not clear.

Any unused product or waste material should be disposed in accordance with local requirements.

MANUFACTURER:

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